

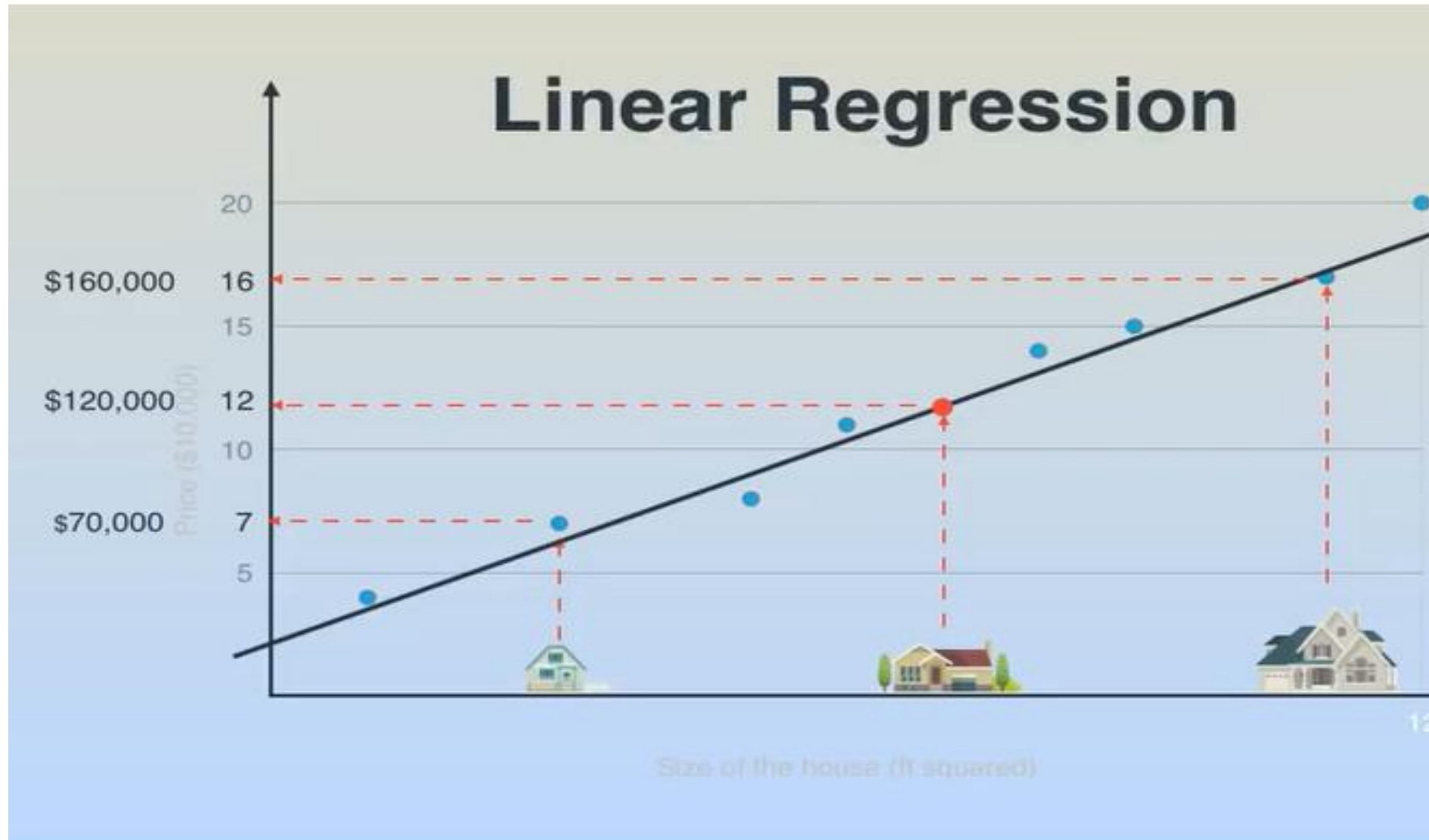
Linear Regression

Introduction

- Linear regression is a powerful and widely used method to estimate **values**, such as
 - price of a house
 - value of a certain stock
 - life expectancy of an individual
 - amount of time a user will watch a video or spend on a website

Linear Regression Intuition

House price estimate – Linear regression

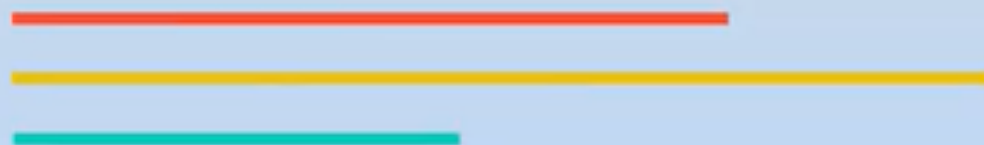


Linear Regression



Gradient descent
(Least squares)

Error:



Gradient descent



Mount Errorest



Linear and Polynomial Regression



Introduction

- **Regression models** describe the **relationship between variables** by fitting a line to the observed data.
- Linear regression models use a **straight line**, while **nonlinear regression models** use a curved line.
- Regression allows you to estimate how a **dependent variable** changes as **the independent variable(s)** change.

Simple Linear Regression

- **Simple linear regression** is used to estimate the relationship between **two quantitative variables**.
- You can use simple linear regression when you want to know:
 - **The relationship is between two variables**
 - e.g. the relationship between salary and experience.
 - The value of the dependent variable at a certain value of the independent variable
 - e.g. the salary at a certain level of experience.

Experience (years) (IDV)	Salary (DV)
1.1	39,343
1.3	46,205
1.5	37,731
2	43,525
2.2	39,891
2.9	56,642
3	60,150

How to perform a simple linear regression?

$$y = b_0 + b_1 * x_1$$

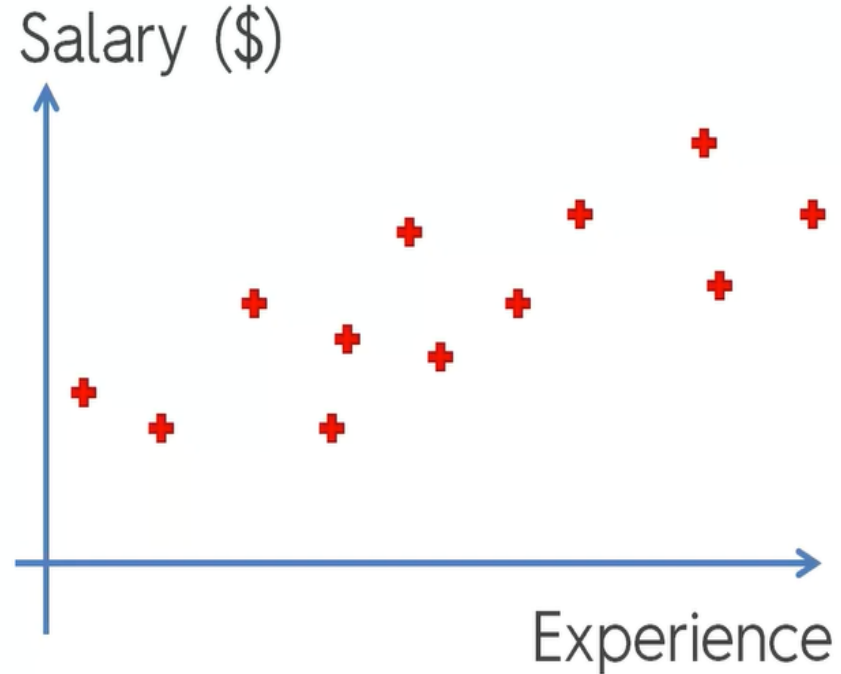
$$y = b_0 + b_1 * x_1$$



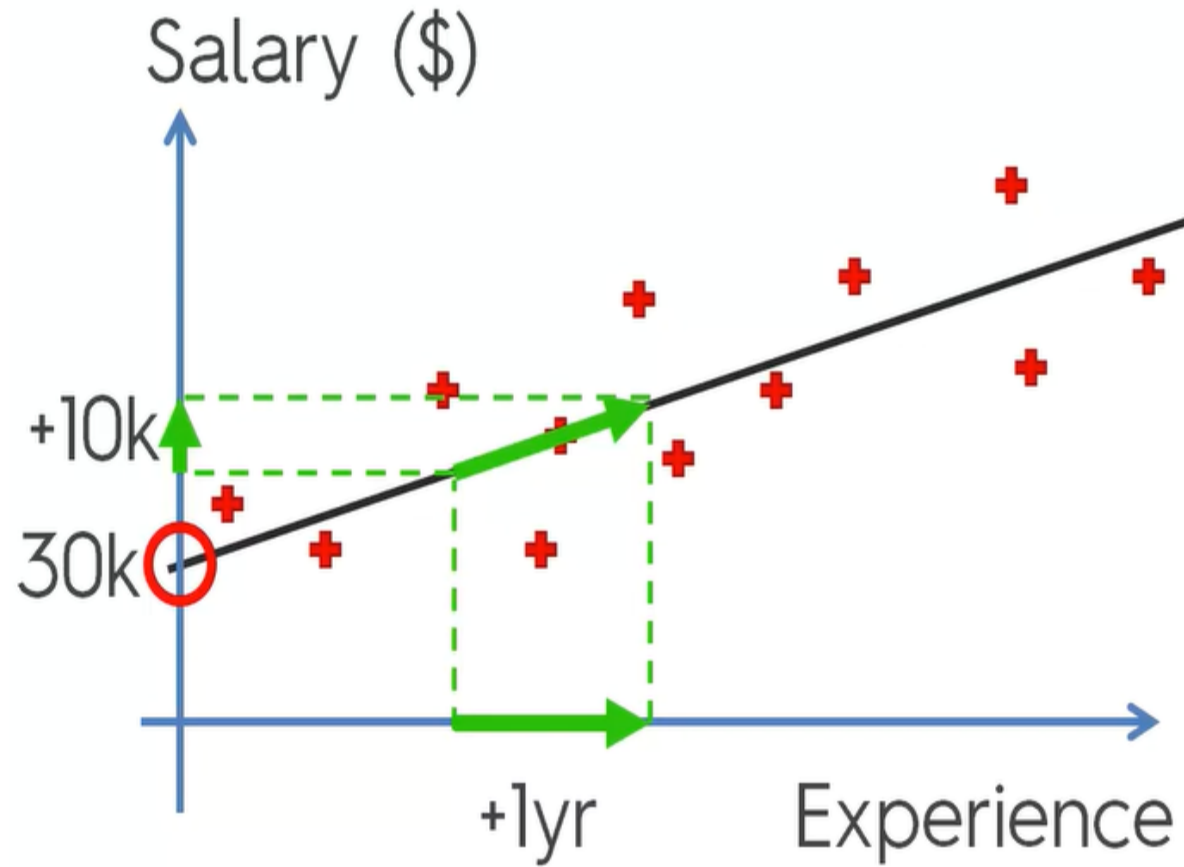
Dependent variable (DV)



Independent variable (IV)



Simple Linear Regression:



$$y = b_0 + b_1 * x$$



$$\text{Salary} = b_0 + b_1 * \text{Experience}$$

slope or
regression
coefficient

$$\hat{y} = b_0 + b_1 * x_1$$

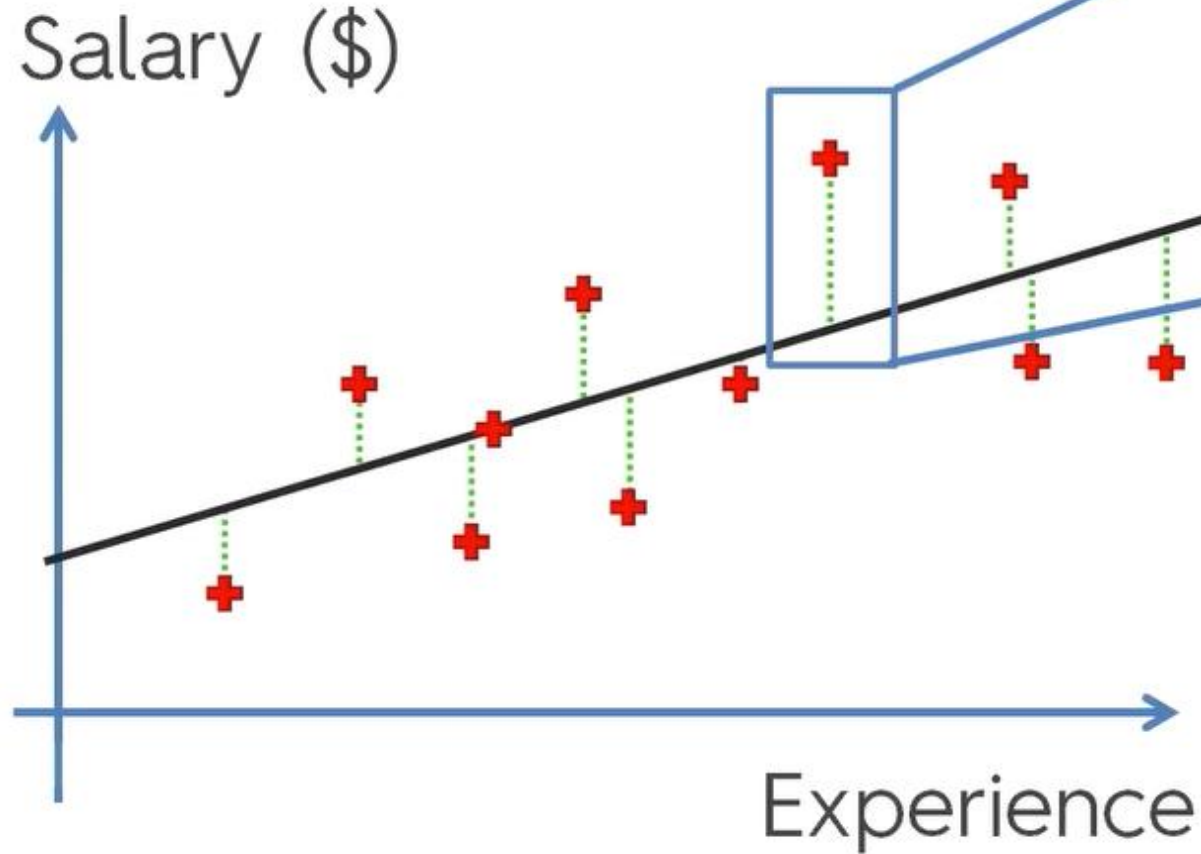
$$\textit{Salary} = b_0 + b_1 * \textit{Experience}$$

$$\textit{Salary} = 30k + 10k * 4$$

$$\textit{Salary} = 70k$$

Experience (years) (IDV)	Salary (DV)
1.1	39,343
1.3	46,205
1.5	37,731
2	43,525
2.2	39,891
2.9	56,642
3	60,150
:	:
:	:

Simple Linear Regression:



$$\text{SUM } (y - \hat{y})^2 \rightarrow \min$$

Study hours and marks

Student	Study hours (x)	Marks (y)
1	1	35
2	2	40
3	3	50
4	4	55
5	5	70

Linear regression finds the line that **minimizes the differences** between the actual marks and predicted marks.

Best-fit line and residuals

residual or prediction error

$$e_i = (y_i - \hat{y}_i)$$

- Positive residual: actual value is higher than the prediction.
- Negative residual: actual value is lower than the prediction.
- Zero residual: prediction is exactly correct.

Sum of square error (SSE)

- Linear regression normally uses the least-squares method.

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- The best-fit line is the line that produces the smallest sum of squared errors.
- Squaring the errors:
 - Prevents positive and negative errors from cancelling.
 - Penalizes large errors more strongly.
 - Produces a function that can be mathematically optimized.

Calculating slope and intercept

- The slope is calculated as:

$$\mathbf{b_1} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

The intercept is: $\mathbf{b_0 = \bar{y} - b_1\bar{x}}$

Example

(x): Study hours	(y): Marks
1	40
2	50
3	50
4	60
5	70
$\bar{x} = \frac{\sum x}{n} = 3$	$\bar{y} = \frac{\sum y}{n} = 54$

Calculate the means

$$\bar{x} = \frac{1 + 2 + 3 + 4 + 5}{5} = 3$$

$$\bar{y} = \frac{40 + 50 + 50 + 60 + 70}{5} = 54$$

$$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Prepare the calculation table

(x)	(y)	$(x - \bar{x})$	$(y - \bar{y})$	$((x - \bar{x}) * (y - \bar{y}))$	$(x - \bar{x})^2$
1	40	-2	-14	28	4
2	50	-1	-4	4	1
3	50	0	-4	0	0
4	60	1	6	6	1
5	70	2	16	32	4
Total				70	10

- Calculate the slope $b_1 = \frac{70}{10} = 7$

- Calculate the intercept $b_0 = 54 - (7 \times 3) = 33$

$$\hat{y} = b_0 + b_1 * x$$

$$\hat{y} = b_0 + b_1 * x$$

Regression equation: $\hat{y} = 33 + 7x$

- Make a prediction: if a student studies for 6 hours:

$$\hat{y} = 33 + 7 * 6 = 75$$

- Calculating residuals

$$\hat{y} = 33 + 7x$$

(x)	Actual (y)	Predicted ($\hat{y}$)	Residual ($y - \hat{y}$)	Squared error ($y - \hat{y}$) ²
1	40	40	0	0
2	50	47	3	9
3	50	54	-4	16
4	60	61	-1	1
5	70	68	2	4
Total				30

$$SSE=30$$

Notice that the residuals add up to zero: $0+3-4-1+2=0$

Evaluation metrics

- Mean Absolute Error = 2
 - Predictions differ from actual marks by approximately two marks on average
- Mean Squared Error = 6
 - MSE penalizes large errors more heavily than MAE.
- Root Mean Squared Error = 2.4
 - RMSE is expressed in the same unit as the target variable.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

- **Coefficient of determination:** Approximately 94.23% of the variation in marks is explained by study hours

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$R^2 = 0.9423$$

A high R^2 does not automatically mean:

- The model is correct.
- The relationship is causal.
- Predictions will be accurate on new data.
- All assumptions are satisfied.

<u>Study Hrs</u> (x)	<u>marks</u> (y)
1	40
2	50
3	50
4	60
5	70

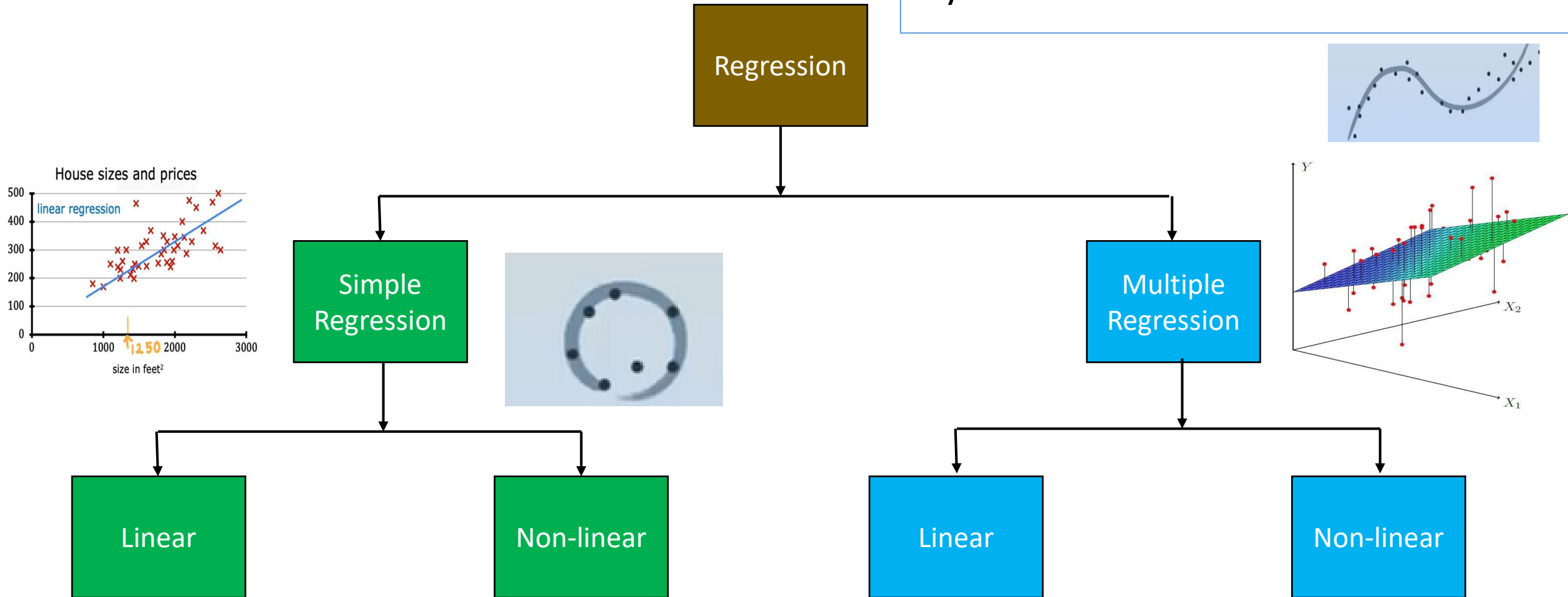
Calculate standard error

$$\text{standard error} = \sqrt{\frac{\sum(\hat{y} - y)^2}{n - 2}}$$

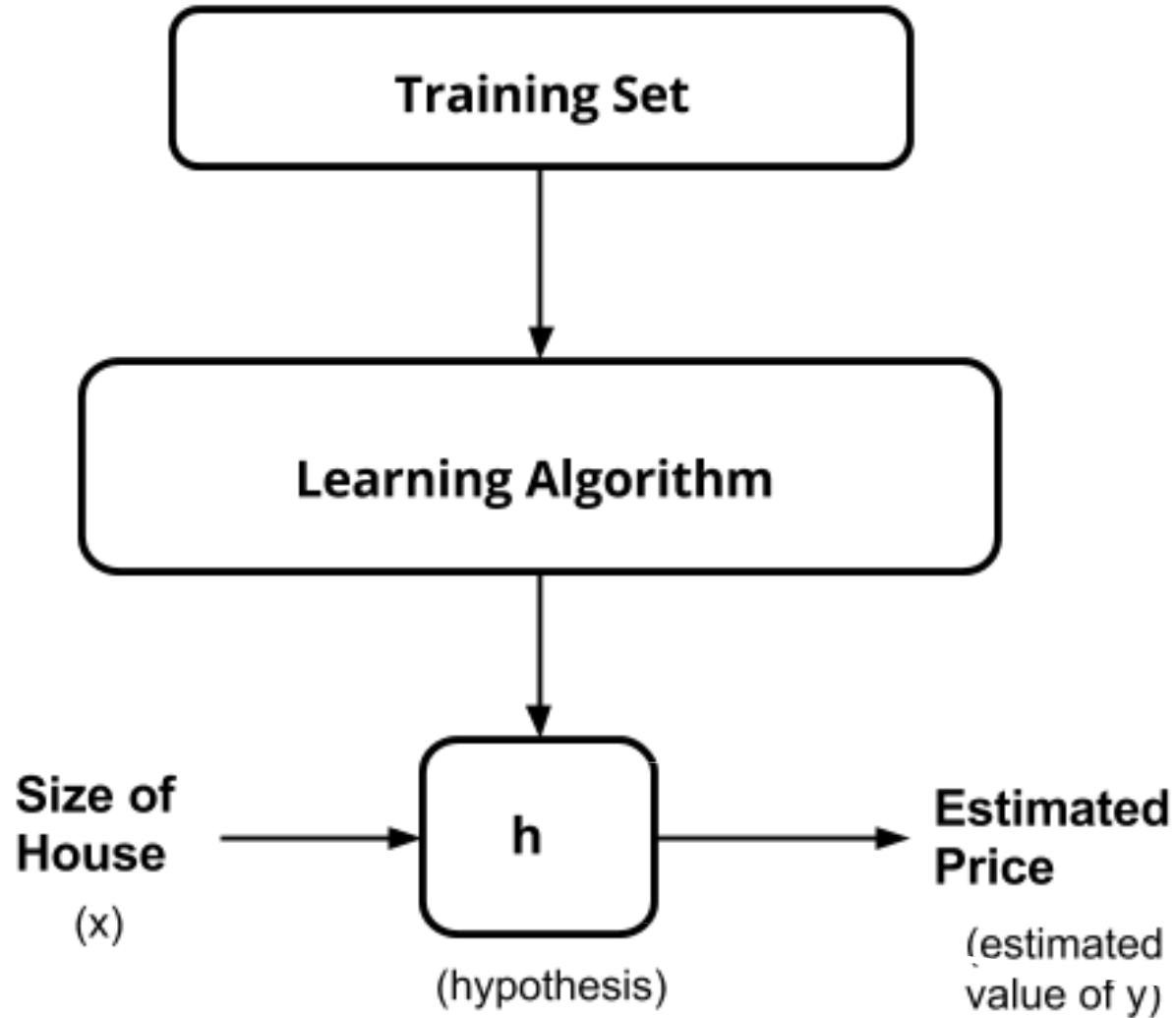
$$\text{standard error} = \sqrt{\frac{30}{5 - 2}} = 3.1622$$

Regression

- It is a supervised learning approach
- If x is given, you can predict $y = f(x)$
- y is continuous value.



Hypothesis function(s)



Linear Regression algorithm

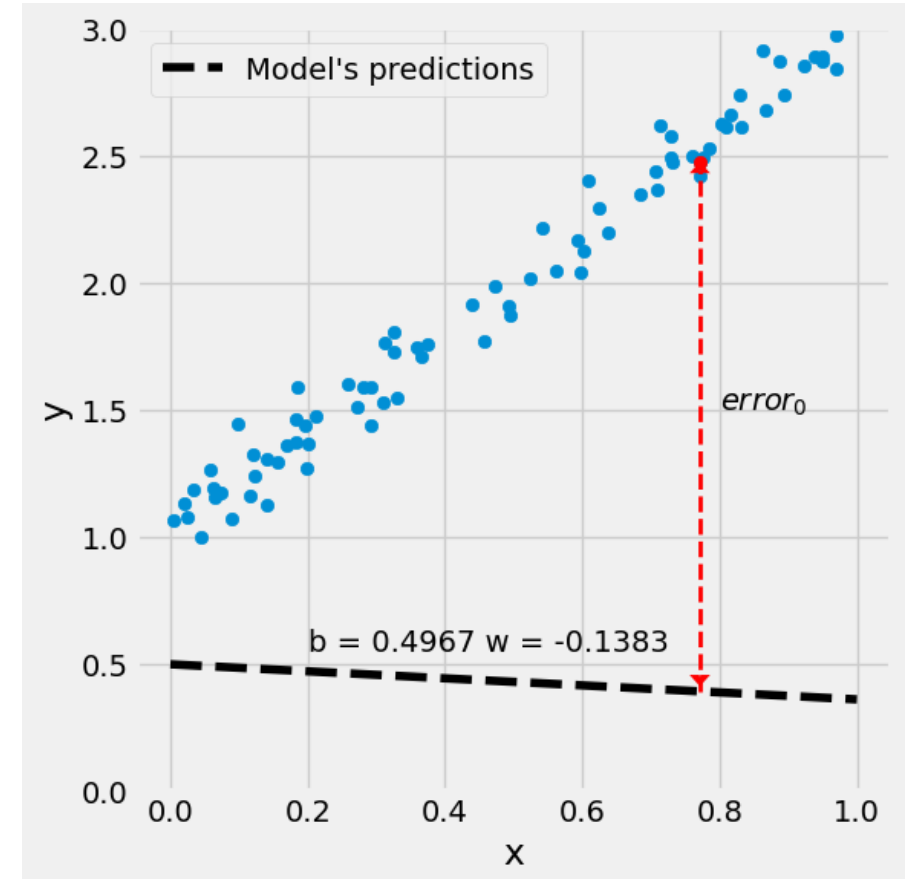
Step 1. Initialization

- Assume parametric form $\hat{y} = wx + b$

Parametric
form of the
hypothesis
function

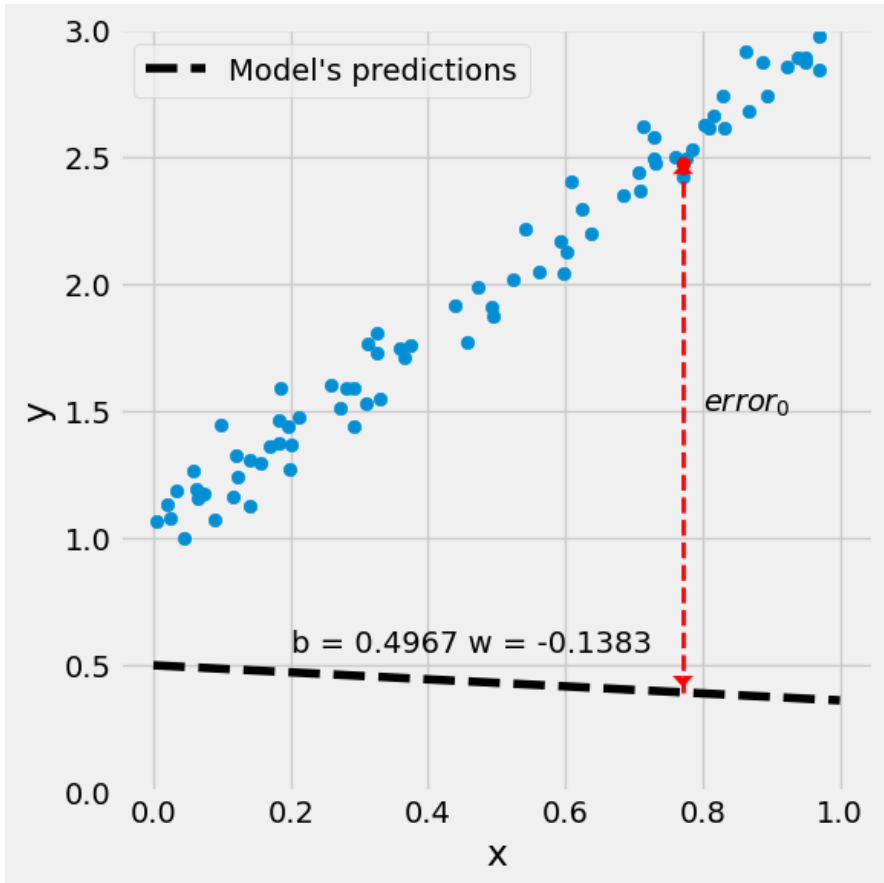
- Calculate error using the initial w & b

$$error^{(i)} = \hat{y}^{(i)} - y^{(i)}$$



Formulate Objective function

Also called
cost / loss
function



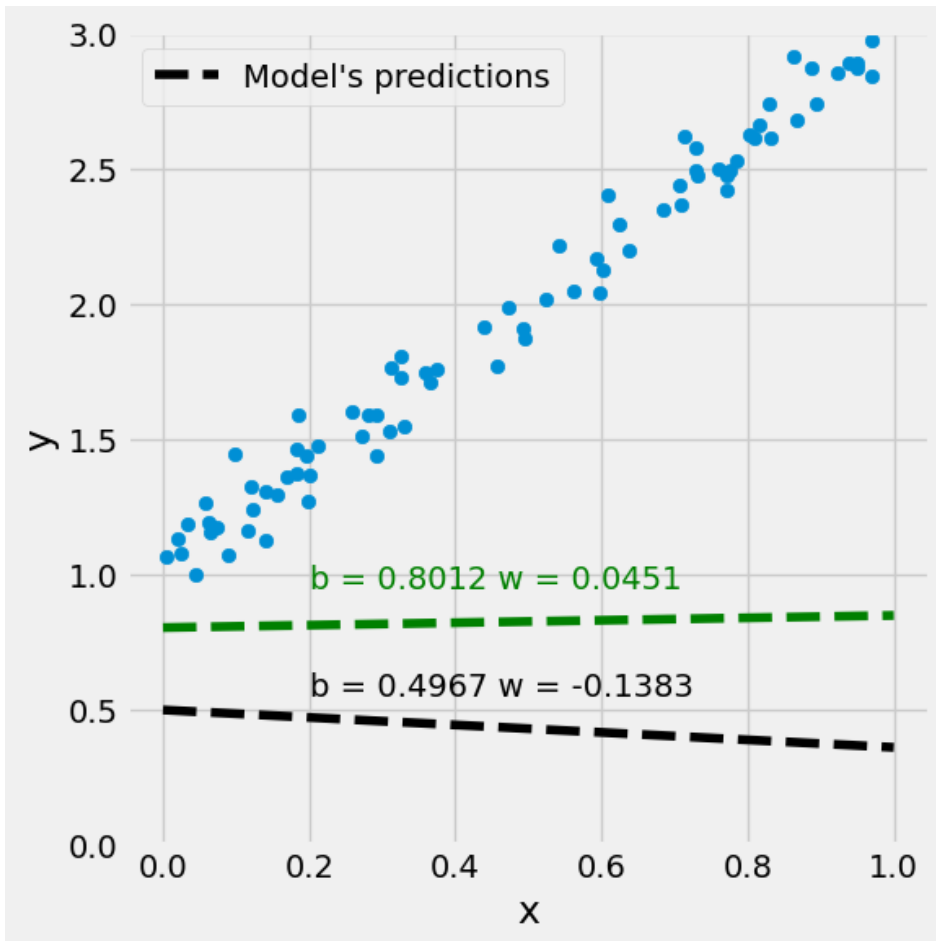
$$error^{(i)} = \hat{y}^{(i)} - y^{(i)}$$

$$MSE = \frac{1}{n} \sum_{i=1}^n error^{(i)2}$$

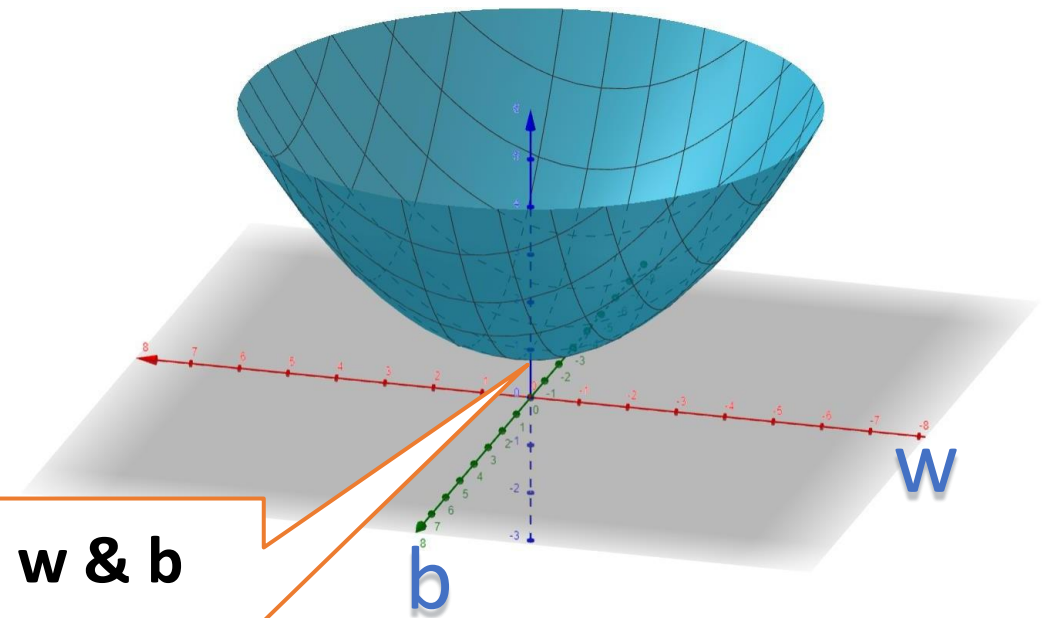
$$= \frac{1}{n} \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)})^2$$

$$\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (b + wx^{(i)} - y^{(i)})^2$$

Feature space versus parameter space

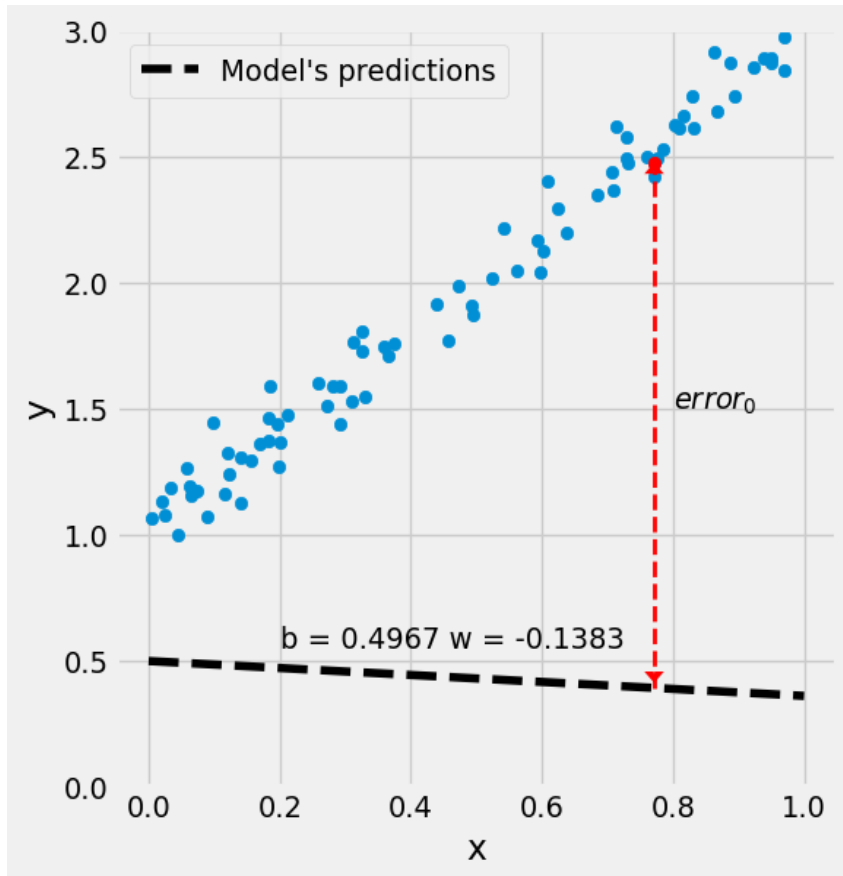


$$\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (b + wx^{(i)} - y^{(i)})^2$$



**Find the w & b
when the cost is
minimum**

Step 2. Evaluate y-hat & evaluate objective function

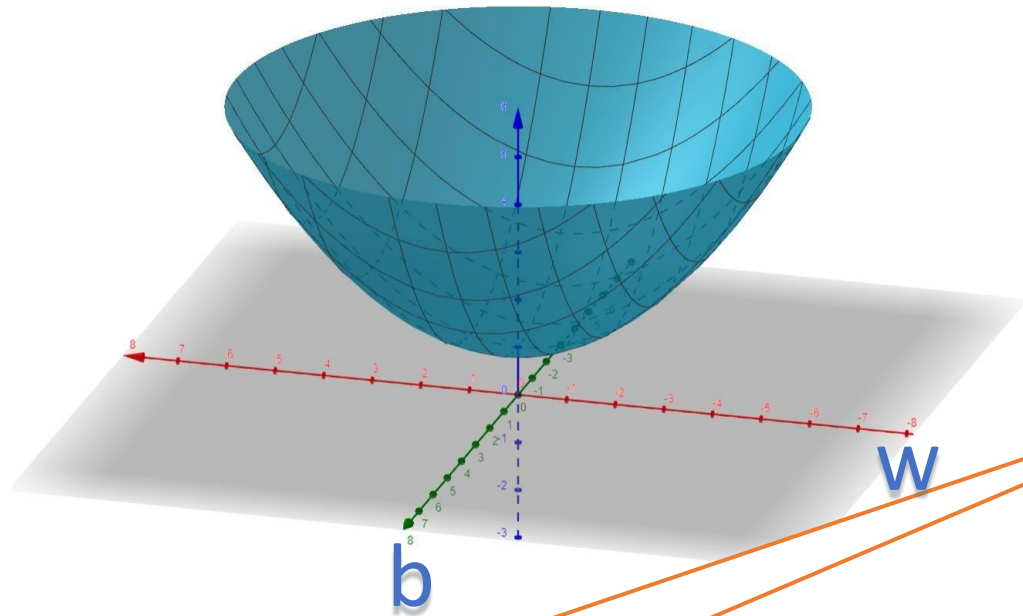


- Evaluate y-hat $\hat{y} = wx + b$
- Evaluate objective function (also known as forward pass)

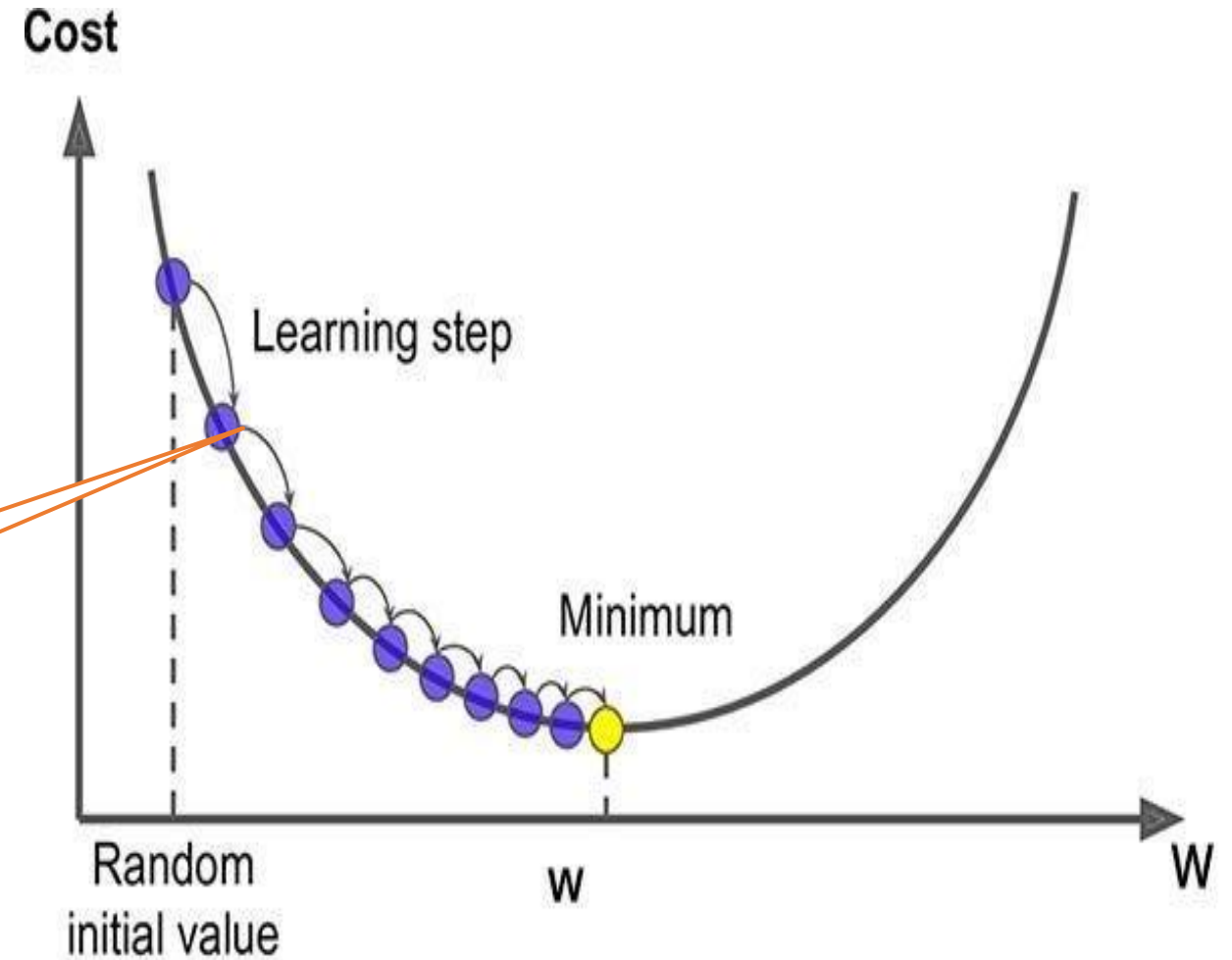
$$\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)})^2$$

$$\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (b + wx^{(i)} - y^{(i)})^2$$

Objective function plot



Take a little step
towards lowest cost.
1. How little?
2. Which direction?



Step 3. Calculate analytical gradients

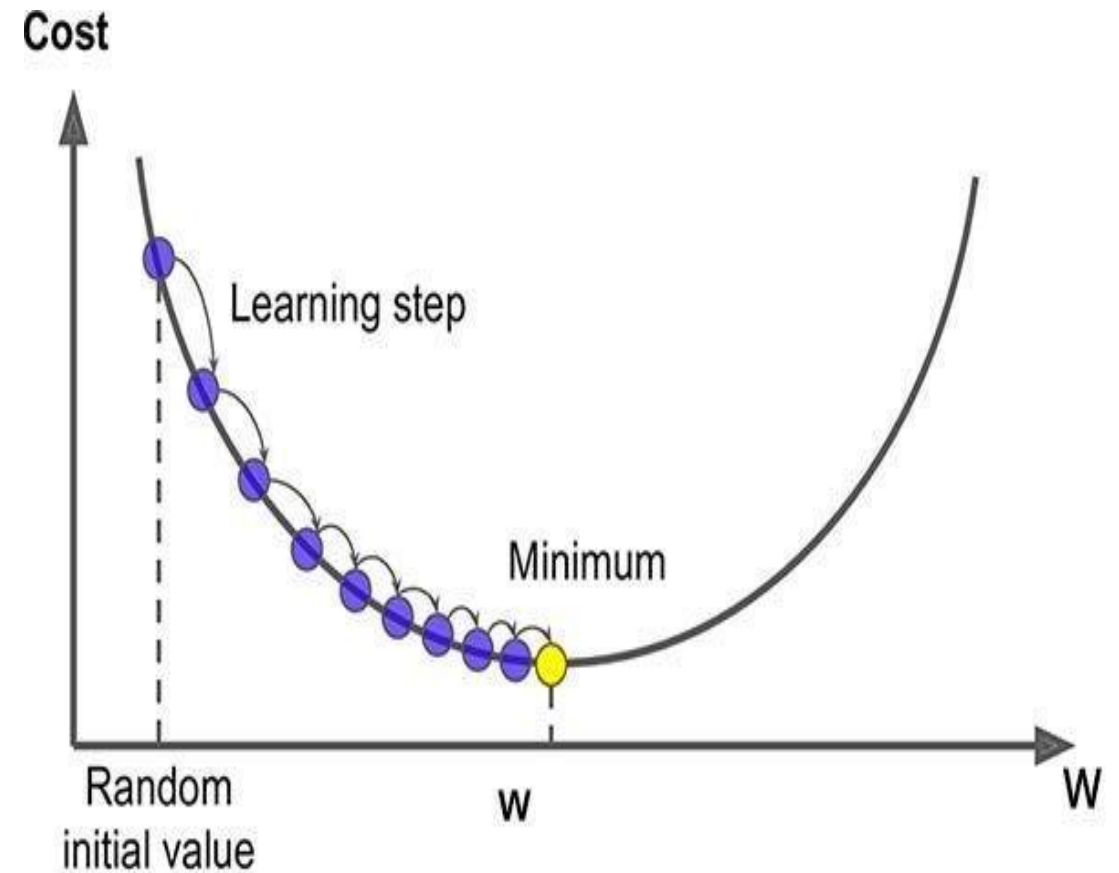
$$\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (b + wx^{(i)} - y^{(i)})^2$$

$$\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (\hat{y}^{(i)} - y^{(i)})^2$$

Calculate gradient

$$\frac{\partial \mathcal{J}}{\partial b} = \frac{1}{m} \sum_{i=1}^m 2(b + wx^{(i)} - y^{(i)})$$

$$\frac{\partial \mathcal{J}}{\partial w} = \frac{1}{m} \sum_{i=1}^m 2x^{(i)}(b + wx^{(i)} - y^{(i)})$$



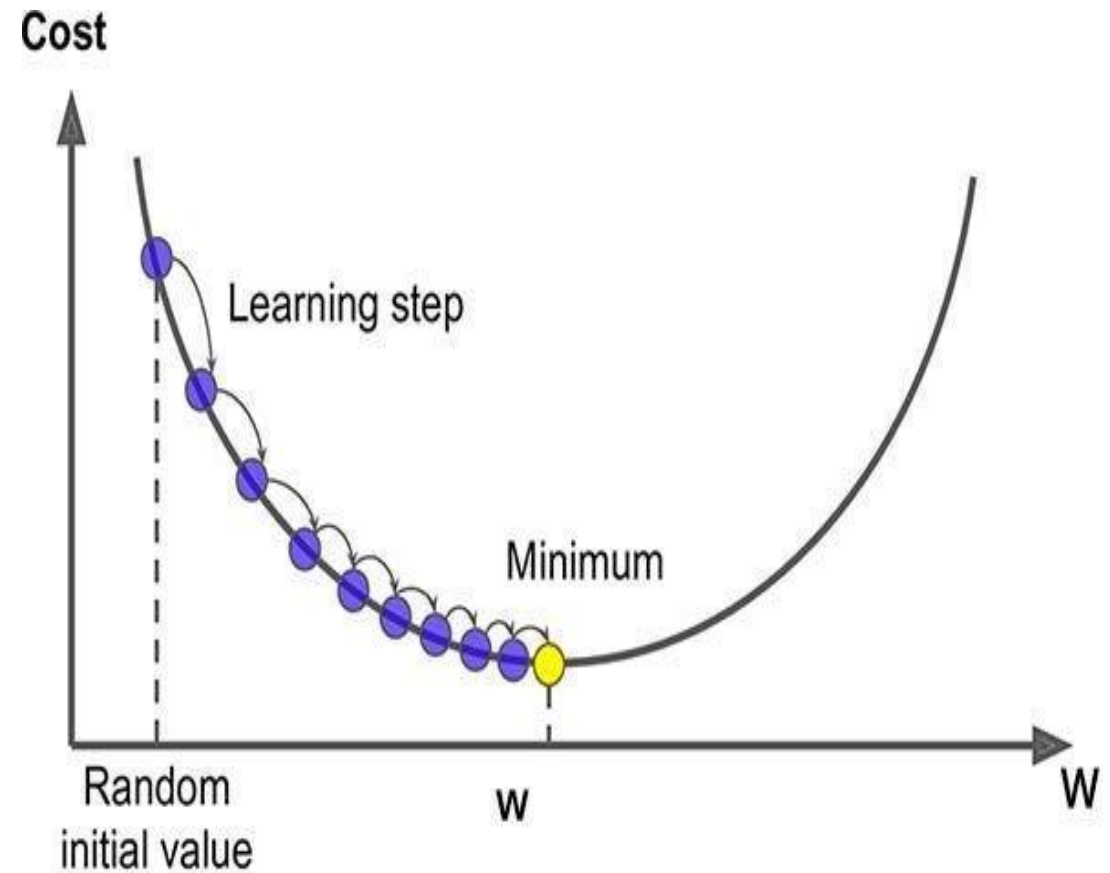
Step 4. Perform numerical gradient descent

- Backward pass – update b and w

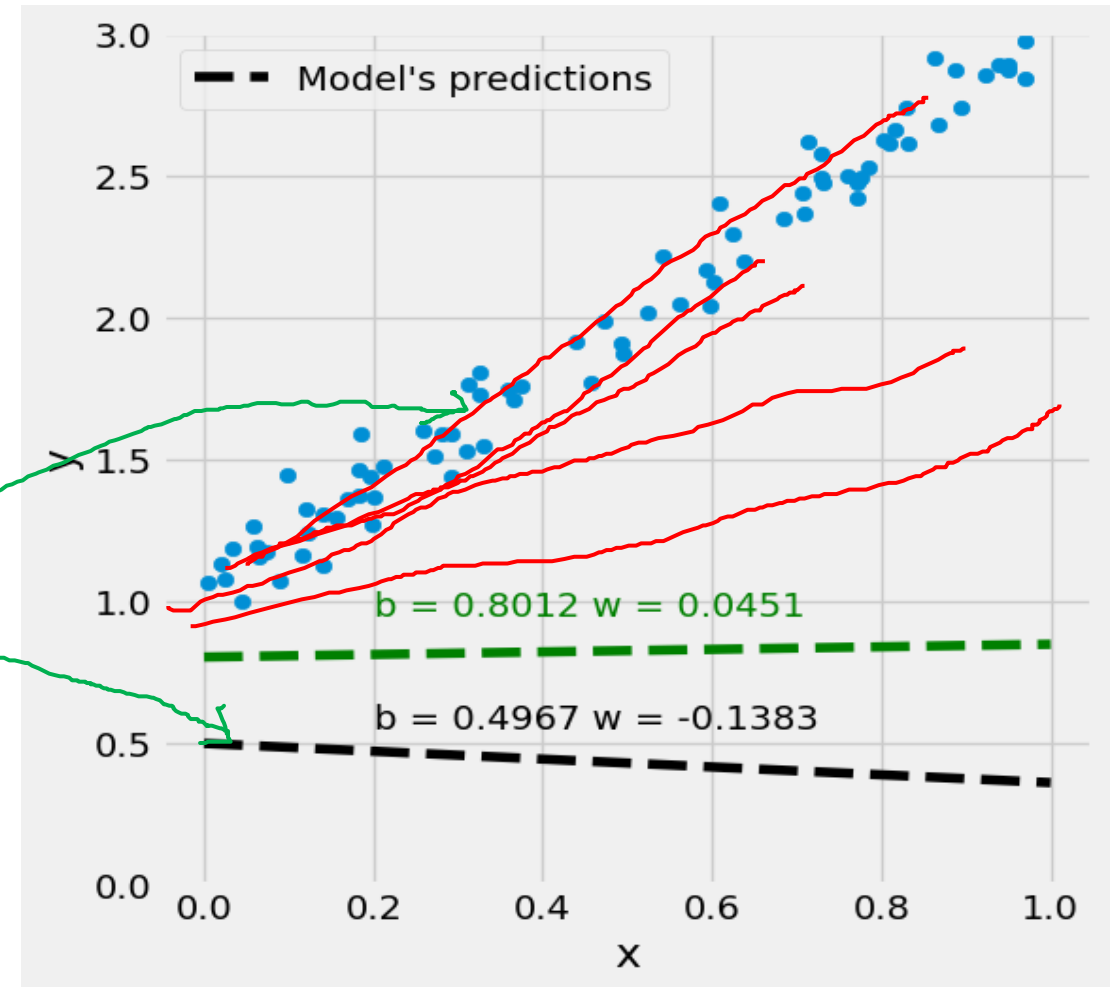
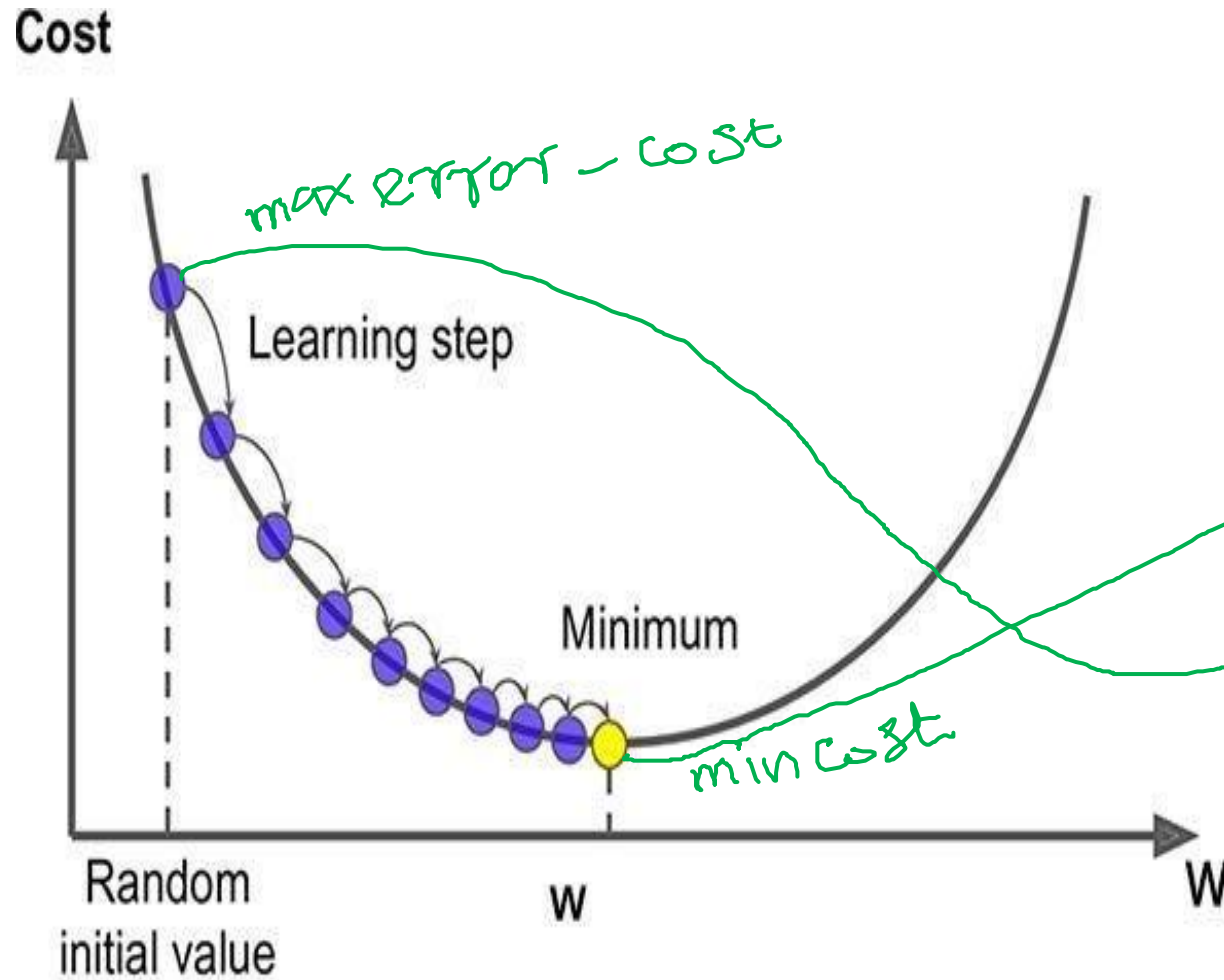
$$b = b - \eta \frac{\partial \mathcal{J}}{\partial b}$$

$$w = w - \eta \frac{\partial \mathcal{J}}{\partial w}$$

- Repeat Step 2, 3, 4



Change in w and b with gradient descent



Review Linear Regression

- Initialization: Select Random w & b , learning rate η

- Loop

- Calculate new-cost for given w & b $\mathcal{J}(w, b) = \frac{1}{n} \sum_{i=1}^n (b + wx^{(i)} - y^{(i)})^2$

- If (iteration == max) or ((new-cost - old cost) < threshold) break

- Calculate gradients wrt w and b $\frac{\partial \mathcal{J}}{\partial b} = \frac{1}{m} \sum_{i=1}^m 2(b + wx^{(i)} - y^{(i)})$

$$\frac{\partial \mathcal{J}}{\partial w} = \frac{1}{m} \sum_{i=1}^m 2x^{(i)}(b + wx^{(i)} - y^{(i)})$$

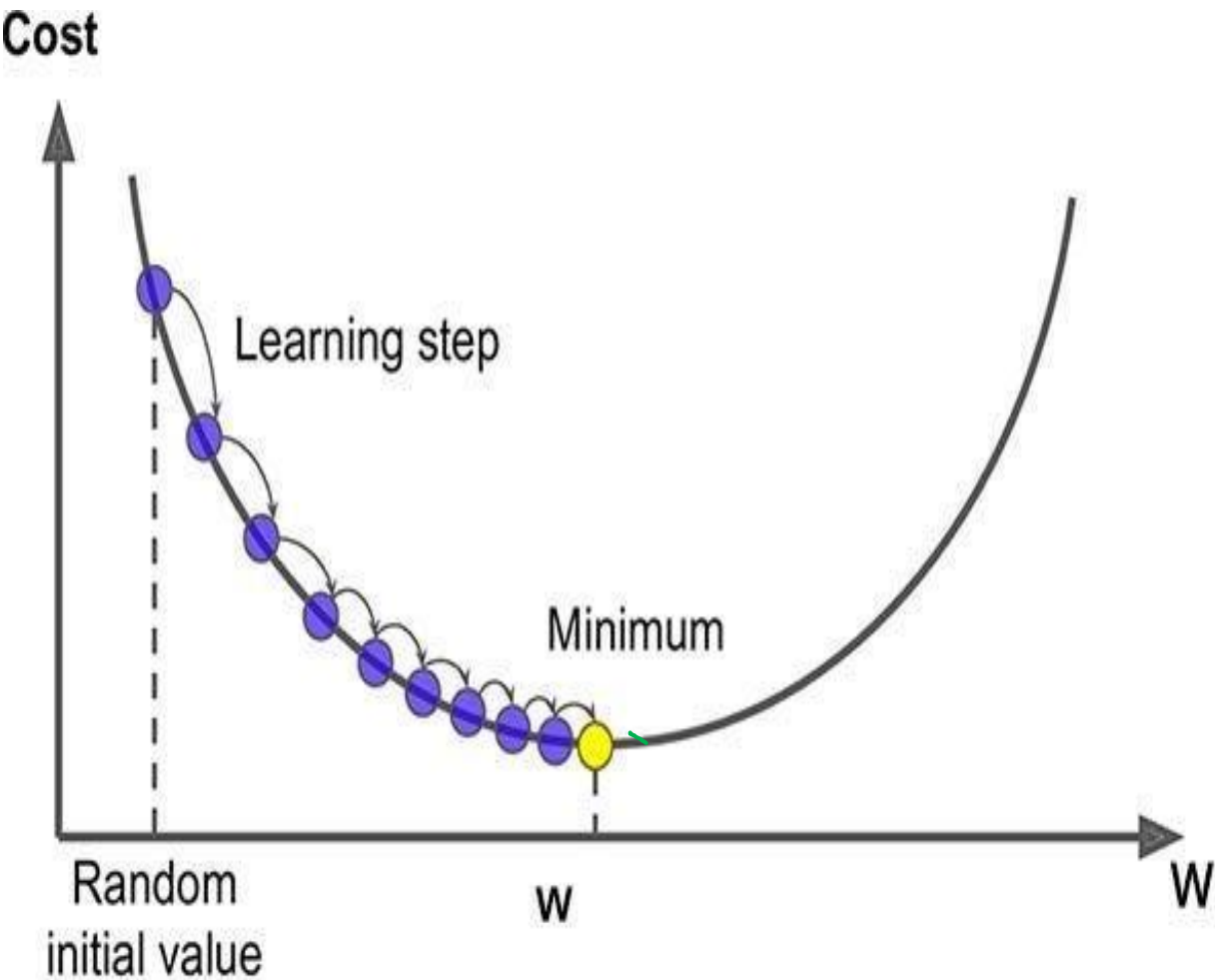
Review Linear Regression

- Do gradient descent to update/ calculate cost function

$$b = b - \eta \frac{\partial \mathcal{J}}{\partial b}$$
$$w = w - \eta \frac{\partial \mathcal{J}}{\partial w}$$

- stop: if old cost = new cost

Gradient descent summary



Iteration	w	b	Cost	dJ/dw	dJ/db
0	0.0110	0.0195	2.0443	-1.1077	-1.9538
1000	1.4309	1.2985	0.0178	-0.0407	0.02078
2000	1.7162	1.1527	0.0071	-0.0191	0.00976
3000	1.8502	1.0842	0.0047	-0.0090	0.00459
4000	1.9132	1.0520	0.0042	-0.0042	0.00215
5000	1.9430	1.0369	0.0041	-0.0020	0.00101
6000	<u>1.9567</u>	<u>1.0298</u>	<u>0.0040</u>	<u>-0.0009</u>	<u>0.00047</u>
7000	1.9632	1.0265	0.0040	-0.0004	0.00022
8000	1.9663	1.0249	0.0040	-0.0002	0.00010
9000	<u>1.9677</u>	<u>1.0242</u>	<u>0.0040</u>	<u>-9.637e-05</u>	<u>4.925e-05</u>

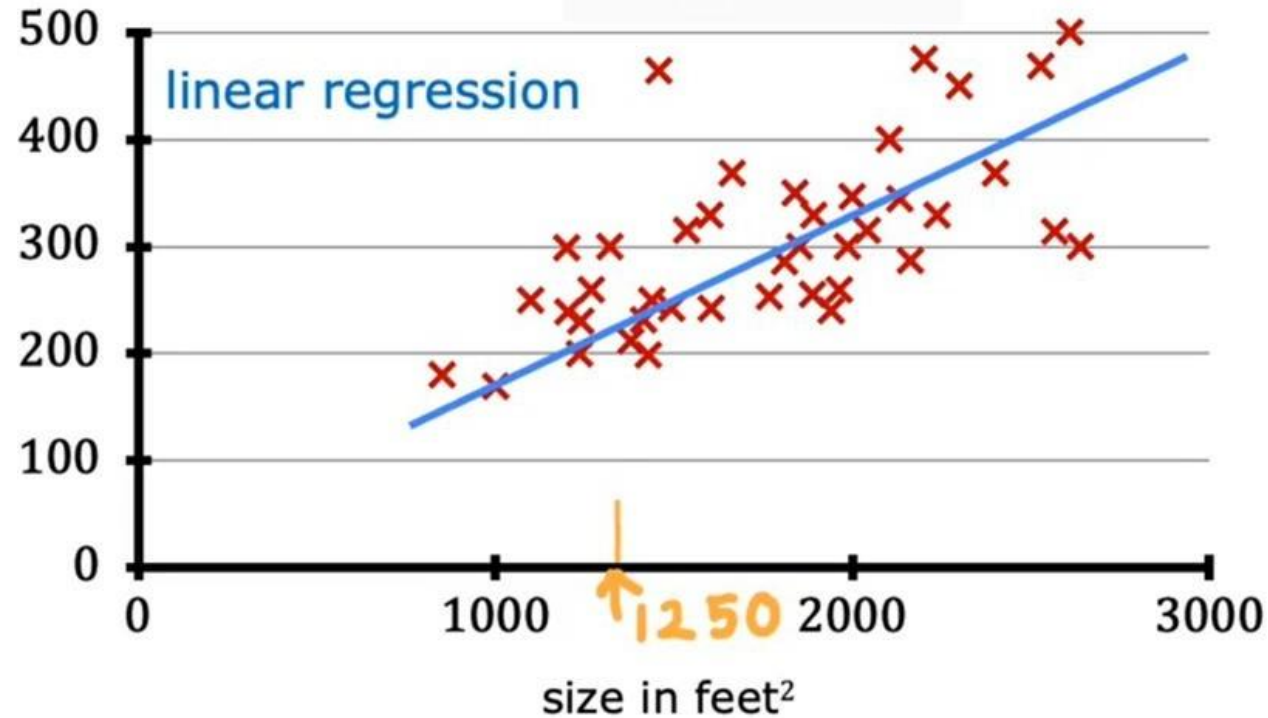
Multiple Linear Regression

Multiple linear regression (MLR)

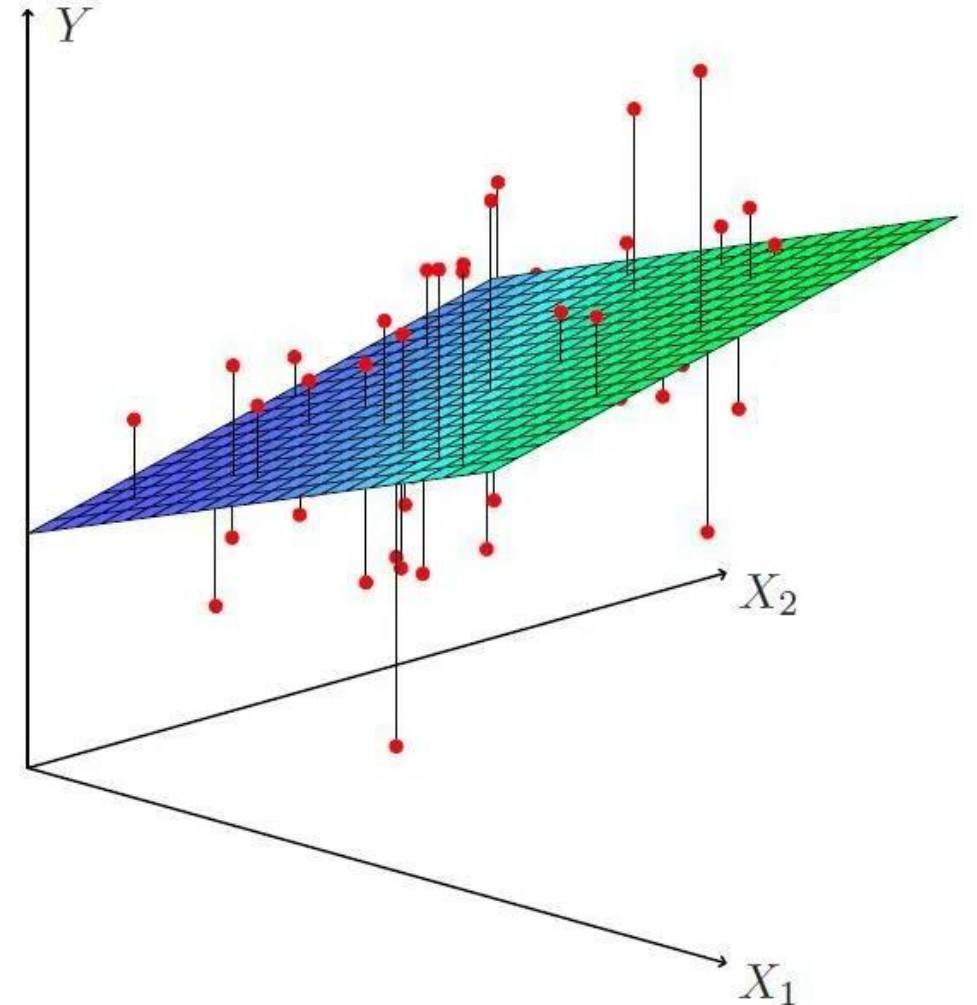
- It is a statistical technique used to predict the relationship between **one dependent variable** and **two or more independent variables**
- Forecasting - Economics, finance
- In biology - patterns detection
- To understand the **impact of multiple elements** on a **singular result**.

Simple Linear Regression

House sizes and prices



Multiple Linear Regression



General Formula

The equation for multiple linear regression is:

$$\hat{y} = b_0 + b_1 * X_1 + b_2 * X_2 + \dots + b_n * X_n + \epsilon$$

b_0 – *y intercept*

b_1, b_2, b_3 – *coefficients*

X_n – *IDV*

$\hat{y}$ – *Predicted DV*

y – *DV*

ϵ – *error term*

Matrix form

Multiple linear regression can be represented as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

The coefficient estimates can be calculated using:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

This is known as the normal equation.

In practical machine-learning projects, libraries such as Scikit-learn calculate the coefficients automatically.

Example

- We want to predict the salary of the person based on the years of experience and education level.

Person	Experience (X_1)	Education (X_2)	Salary (Y)
A	1	12	30,000
B	3	14	40,000
C	5	16	50,000

Example

$$\hat{y} = b_0 + b_1 * Experience + b_2 * Education$$

if $b_0 = 10,000, b_1 = 4,000$ & $b_2 = 1,000$

$$b_0 - \text{starting slaray (10,000)}$$

b_1 – Each additional year of experience increase in salary (4,000)

b_2 – Each additional year of education increase in salary (1,000)

$$\hat{y} = 10,000 + 4,000 * \textit{Experience} + 1,000 * \textit{Education}$$

Example

- Predict the salary of a person who joins the company with 4 years of experience and 15 years of education.

$$\hat{y} = 10,000 + 4,000 * \underline{4} + 1,000 * \underline{15}$$

$$\hat{y} = \underline{41,000}$$

Assumptions of linear regression

Linear regression relies on the following assumption

- **Linearity**

- There should be an approximately **linear relationship** between the **input** variables and the **target**.

- **Independence**

- The **observations** and their **errors** should be **independent**.

- **Homoscedasticity**

- Residuals should have approximately **constant variance** across predicted values.
- When **error variance changes significantly**, it is called **heteroscedasticity**.

- Calculating residuals

$$\hat{y} = 33 + 7x$$

(x)	Actual (y)	Predicted ($\hat{y}$)	Residual ($y - \hat{y}$)	Squared error ($(y - \hat{y})^2$)
1	40	40	0	0
2	50	47	3	9
3	50	54	-4	16
4	60	61	-1	1
5	70	68	2	4
Total				30

significant

95

constant variance

$$SSE=30$$

Notice that the residuals add up to zero: $0+3-4-1+2=0$

Assumptions of linear regression

- **Normality of residuals**

- **Residuals** should be approximately **normally distributed**, particularly when confidence intervals and hypothesis tests are required.

- **No severe multicollinearity**

- In **multiple regression**, **input variables should not be excessively correlated with each other**.
- For example, a dataset containing both “**age in years**” and “**age in months**” would create severe multicollinearity.

- **No influential outliers**

- **Extreme observations** can significantly change the slope and intercept.

Prepare the calculation table

(x)	(y)	$(x - \bar{x})$	$(y - \bar{y})$	$((x - \bar{x}) * (y - \bar{y}))$	$(x - \bar{x})^2$
1	40	-2	-14	28	4
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4	60	1	6	6	1
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Total				70	10

- Calculating residuals

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Total				30

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Evaluation metrics

- Mean Absolute Error = 2
 - Predictions differ from actual marks by approximately two marks on average
- Mean Squared Error = 6
 - MSE penalizes large errors more heavily than MAE.
- Root Mean Squared Error = 2.4
 - RMSE is expressed in the same unit as the target variable.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE}$$

- **Coefficient of determination:** Approximately 94.23% of the variation in marks is explained by study hours

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

$$R^2 = 0.9423$$

A high R^2 does not automatically mean:

- The model is correct.
- The relationship is causal.
- Predictions will be accurate on new data.
- All assumptions are satisfied.

<u>Study Hrs</u> (x)	<u>marks</u> (y)
1	40
2	50
3	50
4	60
5	70

Adjusted R^2

- Adding unnecessary input variables can increase R^2
- Adjusted R^2 accounts for the number of predictors.

$$R^2_{\text{adj}} = 1 - (1 - R^2) \frac{n - 1}{n - \underline{p} - 1}$$

Where:

- n : number of observations
- p : number of predictors
- Adjusted R^2 is especially **useful in multiple regression**.

A **predictor** is an input variable used by a machine-learning model to predict the output.

Predictors are also called:

- **Features**
- **Independent variables**
- **Input variables**
- **Explanatory variables**

Multicollinearity

- Multicollinearity occurs when **two or more predictors** are **strongly related**.

Consequences include:

- Unstable coefficient estimates
- Unexpected coefficient signs
- Difficulty interpreting individual variables
- Large standard errors

- Consider a model that predicts a student's final mark using three predictors:

Student	Study hours/day X_1	Study hours/week X_2	Attendance X_3	Final mark Y
A	1	7	60	45
B	2	14	70	55
C	3	21	75	65
D	4	28	85	75
E	5	35	90	85

- Study hours per day and study hours per week provide exactly the same information. (**strongly related**)
- They are perfectly correlated.

$$X_2 = 7X_1$$

- If both are included in the regression model,

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$$

- The model **cannot reliably determine** whether the **improvement in marks is caused by X_1 or X_2** , because they always change together.
- For example, one fitted model might assign: $\beta_1=7, \beta_2=0$
- Another fitting method might assign: $\beta_1=0, \beta_2=1$
- Both produce the same contribution because $X_2 = 7X_1$.
- This makes the individual coefficients **unstable and difficult to interpret**.

- A practical solution is to **retain only one of the two variables**:

$$\hat{Y} = \beta_0 + \beta_1(\text{Study hours/day}) + \beta_2(\text{Attendance})$$

- Multicollinearity occurs when two or more predictors contain the same or nearly the same information.

Other common examples include:

- Age in years and age in months
- Temperature in Celsius and Fahrenheit
- Distance in kilometres and metres
- Monthly income and annual income
- House area in square feet and square metres

How to detect multicollinearity

- The **Variance Inflation Factor** can be used to detect it:

$$VIF_j = \frac{1}{1 - R_j^2}$$

General guidelines:

- $VIF = 1$: no multicollinearity
- $VIF > 5$: possible concern
- $VIF > 10$: serious concern

These are guidelines, not absolute rules.

Underfitting and overfitting

Underfitting

- The model is too simple to capture the relationship.

Indicators:

- High training error
- High testing error
- Systematic patterns in residuals

Overfitting

- The model learns training data too closely and performs poorly on new data.

Indicators:

- Very low training error
- High testing error
- Too many irrelevant input variables

Possible solutions:

- Collect more data.
- Remove irrelevant features.
- Use cross-validation.
- Apply Ridge or Lasso regression.
- Reduce model complexity.

Linear, Ridge, and Lasso regression

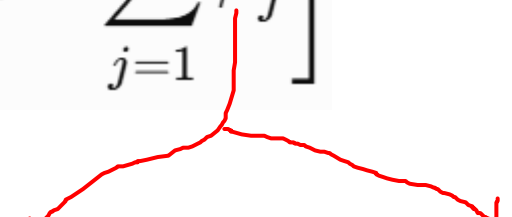
- **Linear regression:** Minimizes the prediction error.

$$\min \sum (y_i - \hat{y}_i)^2$$

- **Ridge regression:** Ridge adds an L2 penalty.

- Ridge reduces coefficient magnitudes but usually does not make them exactly zero.

$$\min \left[\sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right]$$


$$\hat{y} = \underline{b_0} + \underline{b_1} * \textit{Experience} + \underline{b_2} * \textit{Education}$$

Prediction error

- Ridge regression searches for **coefficient values** that produce a **good prediction** while also keeping the **coefficients small**.
- The two parts of the objective function

$$\min \left[\sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right]$$

Least-squares error term

L2 penalty

This is the ordinary least-squares error.
Minimizing it encourages the regression model to **fit the training data closely**.

(2) L2 penalty

$$\lambda \sum_{j=1}^p \beta_j^2$$

where:

- β_j is the coefficient of predictor X_j .
- p is the number of predictors.
- λ controls the strength of the penalty.

$\sum \beta_j^2$ is called the **L2 penalty** or **squared L2 norm**.

$$\beta_1^2 + \beta_2^2 + \dots + \beta_p^2$$

It is called L2 because the coefficients are squared:
Usually, the intercept β_0 is **not penalized**.

Why is a penalty required?

- Ordinary linear regression **minimizes** only the **prediction error**:

$$\min_{\beta} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- It may sometimes **produce very large coefficients**, particularly when:
 - Predictors are strongly correlated.
 - There are many predictors.
 - The dataset is small.
 - The model is overfitting.
 - Small changes in the data cause large coefficient changes.

Ridge regression tells the model:

- **Fit the data well**, but **avoid** using unnecessarily **large coefficients**.
- Therefore, the model must balance two goals:

Small prediction error + Small coefficients

How the L2 penalty changes model selection

- Suppose two candidate regression models produce the following results.

Model A:

- SSE = 20
- Coefficients: $\beta_1=10$, $\beta_2=8$
- The L2 penalty is:

$$\beta_1^2 + \beta_2^2 = 10^2 + 8^2 = 164$$

- If $\lambda = 1$, the total objective is:
 $20+1(164)=184$

Model B:

$$\text{SSE}=25$$

$$\text{Coefficients: } \beta_1=2, \beta_2=1$$

- The L2 penalty is:
- If $\lambda = 1$, the total objective is:
 $25+1(5)=30$

Although Model A has the smaller training error, Ridge regression **selects Model B** because it has much smaller coefficients and a much smaller total objective: **30 < 184**

Ridge regression may accept a **small increase in training error** to obtain a more stable and less complex model.

Why are the coefficients squared?

- Squaring has three important effects.
 - Large coefficients receive stronger penalties
 - Positive and negative coefficients are treated equally
 - The penalty is smooth
- **Large coefficients receive stronger penalties**

Coefficient β	Squared penalty β^2
0.5	0.25
1	1
2	4
5	25
10	100

Increasing a coefficient from 1 to 2 increases the penalty from 1 to 4. Increasing it from 5 to 10 increases the penalty from 25 to 100.

Thus, very large coefficients become expensive.

Positive and negative coefficients are treated equally

$$(-5)^2 = 5^2 = 25$$

- Therefore, Ridge penalizes the **magnitude** of a coefficient, not its direction.

The penalty is smooth:

- The derivative of the penalty for coefficient β_j is:

$$\frac{\partial}{\partial \beta_j} (\lambda \beta_j^2) = 2\lambda \beta_j$$

Role of λ

- The parameter λ , determines how strongly the coefficients are penalized.

When $\lambda = 0$

$$SSE + 0 \sum \beta_j^2 = SSE$$

- There is **no penalty**. Ridge regression becomes **ordinary linear regression**.

When λ is small

- Prediction error receives more importance. The coefficients are reduced only slightly.

Role of λ

When λ is large

- Keeping coefficients small becomes more important. The coefficients shrink considerably toward zero.

When λ becomes extremely large

- The slope coefficients approach zero:
- The model becomes very simple and may underfit.

$$\beta_1, \beta_2, \dots, \beta_p \rightarrow 0$$

Value of λ	Coefficient size	Likely effect
0	Original OLS coefficients	No regularization
Small	Slightly reduced	Mild regularization
Moderate	Noticeably reduced	May improve generalization
Very large	Close to zero	Possible underfitting

The best λ is normally selected using **cross-validation**.

Predictor	$\lambda = 0$	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$
Study hours	4.20	4.10	3.60	1.20
Attendance	0.28	0.26	0.15	0.13
Assignment score	0.15	0.13	0.12	0.11
Student ID	0.03	0.01	0.002	0.001

These values are illustrative, but they show the typical coefficient-shrinkage pattern.

How Ridge handles multicollinearity

- Consider predicting marks using:
- X_1 : study hours per day
- X_2 : study hours per week
- These predictors are strongly related because:

$$X_2 \approx 7X_1$$

Student	Study hours/day X_1	Study hours/week X_2	Attendance X_3	Final mark Y
A	1	7	60	45
B	2	14	70	55
C	3	21	75	65
D	4	28	85	75
E	5	35	90	85

- Ordinary regression may produce unstable coefficients, such as:

$$\hat{y} = 30 + 12X_1 - 0.8X_2$$

- A **small change** in the dataset might produce a very different model:

$$\hat{y} = 30 - 2X_1 + 1.2X_2$$

- The predictions may remain similar, but the individual coefficients change considerably.
- Ridge regression **penalizes large coefficients** and tends to distribute the predictive contribution more smoothly between correlated predictors.

$$\hat{y} = 30 + 2.5X_1 + 0.4X_2$$

Ridge provides:

- Smaller coefficients
- More stable coefficient estimates
- Reduced sensitivity to small data changes
- Better performance on unseen (test) data in many cases
- However, Ridge does not remove the underlying correlation. It reduces its harmful effect on coefficient estimation.

- For ordinary multiple linear regression, the coefficient estimate is:

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

- When **predictors are highly correlated**, $\mathbf{X}^T \mathbf{X}$ can be nearly singular.
- Its **inverse becomes unstable**, leading to **large** or **unreliable coefficients**.
- Ridge regression modifies the equation:

$$\hat{\beta}_{\text{Ridge}} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

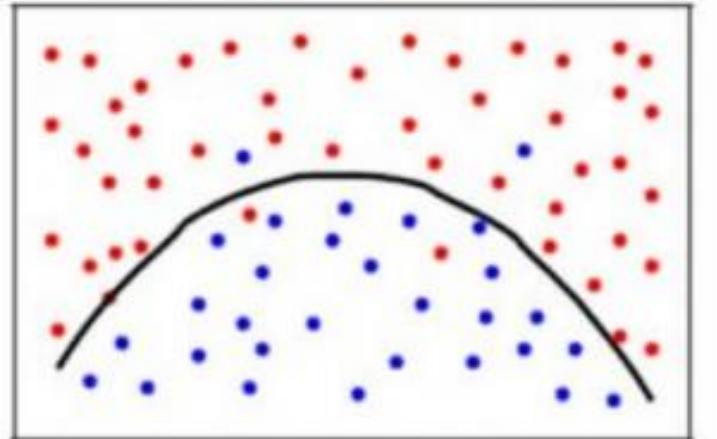
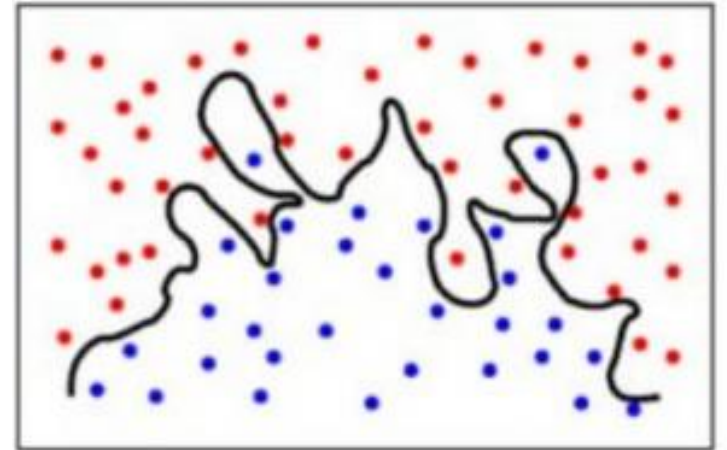
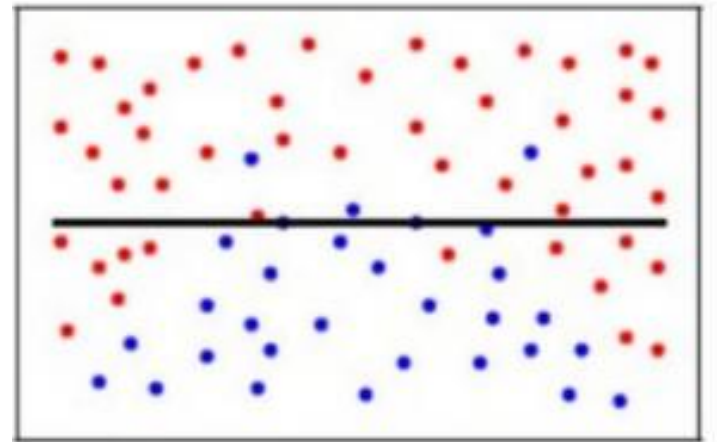
- $\mathbf{I}$ is the identity matrix.
- $\lambda \mathbf{I}$ adds a positive quantity to the diagonal of $\mathbf{X}^T \mathbf{X}$.
- This makes the matrix inversion more stable, particularly when the predictors are strongly correlated.

Bias–variance trade-off

- Ridge regression intentionally introduces a small amount of bias by shrinking coefficients. In return, it can substantially reduce variance.
- **Ordinary regression:** lower bias but possibly high variance.
- **Ridge regression:** slightly higher bias but usually lower variance.
- A less exact fit to the training data can sometimes provide better predictions on test data.

Small increase in bias $\rightarrow$ Large reduction in variance

- This is why Ridge regression can reduce overfitting.



Why feature scaling is important

Suppose the predictors are:

- **Age:** approximately 20–60
- **Annual income:** approximately 300,000–2,000,000
- Because the L2 penalty acts directly on coefficient values, features measured on different scales may be penalized unfairly.
- Therefore, predictors should generally be standardized before applying Ridge regression:

After standardization, each predictor has approximately:

- Mean = 0
- Standard deviation = 1

$$z = \frac{x - \mu}{\sigma}$$

This allows the penalty to treat predictors more fairly.

Does Ridge make coefficients exactly zero?

- Usually, no.

Ridge regression: $\lambda \sum \beta_j^2$

shrinks coefficients toward zero, but generally does not make them exactly zero.

For example: $5.2 \rightarrow 2.1 \rightarrow 0.7 \rightarrow 0.08$

- The coefficient becomes very small but typically remains nonzero.
Therefore, Ridge usually retains all predictors.

- Lasso regression uses an L1 penalty: $\lambda \sum |\beta_j|$

and can make some coefficients exactly zero. Hence:

- **Ridge:** coefficient shrinkage
- **Lasso:** shrinkage and feature selection

L2 penalty discourages large coefficients by adding their squared values to the loss.

Ridge regression does not simply minimize prediction error. It selects coefficients that provide a good balance between **accuracy, simplicity and stability**.

Lasso regression

- Lasso regression: Lasso adds an **L1** penalty to the ordinary least-squares error :
- Lasso can reduce some **coefficients to exactly zero** and it can automatically **select important predictors**.

$$\min \left[\sum (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

- **Example: predicting students' final marks**

Suppose final marks are predicted using four variables:

X_1 : study hours

X_2 : attendance percentage

X_3 : assignment score

X_4 : student **identification number**

The linear regression model is:
$$\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Suppose ordinary linear regression produces:

$$\hat{y} = 8 + 4.2X_1 + 0.28X_2 + 0.15X_3 + 0.03X_4$$

The student **identification number** has **no meaningful relationship** with marks. However, **ordinary regression** has assigned it a **small nonzero coefficient**.

- After applying Lasso regression, suppose the coefficients become:

$$\hat{y} = 10 + 3.8X_1 + 0.25X_2 + 0.12X_3 + 0X_4$$

Since: $\beta_4=0$

- the **identification-number predictor is removed** from the model.
- This is why Lasso performs **feature selection**.

Example:

- Suppose a model has the following coefficients:

$$\beta_1 = 4, \quad \beta_2 = 2, \quad \beta_3 = 0.5$$

The L1 penalty is:

$$\begin{aligned} & |\beta_1| + |\beta_2| + |\beta_3| \\ &= |4| + |2| + |0.5| = 6.5 \end{aligned}$$

If $\lambda=2$

$$\lambda \sum |\beta_j| = 2(6.5) = 13$$

Suppose the model's SSE = 20.

Lasso objective value is: $20+13 = 33$

During training, Lasso searches for coefficient values that **minimize this combined value.**

Comparing two candidate models

Suppose two models produce these results.

Model A

SSE=15

Coefficients:

$$\beta_1 = 5, \quad \beta_2 = 3, \quad \beta_3 = 2$$

For $\lambda = 2$:

$$L1 = |5| + |3| + |2| = 10$$

$$\text{Objective} = 15 + 2(10) = 35$$

Model B

SSE=20

Coefficients:

$$\beta_1 = 3, \quad \beta_2 = 1, \quad \beta_3 = 0$$

For $\lambda = 2$:

$$L1 = |3| + |1| + |0| = 4$$

$$\text{Objective} = 20 + 2(4) = 28$$

Although Model A has the lower training error, Lasso prefers Model B because:

$$28 < 35$$

Model B is simpler and removes the third predictor.

Effect of λ

The value of λ controls the strength of regularization.

Value of λ	Effect			
$\lambda = 0$	Same as ordinary linear regression			
Small λ	Slightly shrinks coefficients			
Moderate λ	Some coefficients may become zero			
Very large λ	Most or all coefficients may become zero			

Predictor	$\lambda = 0$	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$
Study hours	4.20	4.10	3.60	1.20
Attendance	0.28	0.26	0.15	0
Assignment score	0.15	0.12	0	0
Student ID	0.03	0	0	0

These values are illustrative, but they show the typical coefficient-shrinkage pattern.

Ridge versus Lasso

Characteristic	Ridge	Lasso
Penalty	$\lambda \sum \beta_j^2$	$\lambda \sum \beta_j $
Penalty type	L2	L1
Shrinks coefficients	Yes	Yes
Makes coefficients exactly zero	Usually no	Yes
Performs feature selection	No	Yes
Correlated predictors	Usually distributes their contribution	May select one and remove others
Suitable when	Most predictors are useful	Only some predictors are useful

Lasso improves simplicity by shrinking unimportant coefficients exactly to zero.

Short quiz

1. Linear regression predicts:

- A. Categories
 - B. Continuous values
 - C. Clusters
 - D. Association rules
- B. Continuous values

2. Which function is minimized by ordinary least squares?

- A. Sum of residuals
 - B. Sum of absolute input values
 - C. Sum of squared residuals
 - D. Number of features
- C. Sum of squared residuals

3. If the slope is negative:

- A. y generally decreases as x increases.
- B. y always becomes zero.
- C. x and y are unrelated.
- D. The model is invalid.

A. y generally decreases as x increases.

4. Which metric is in the same unit as the output?

- A. MSE only
- B. RMSE
- C. R^2
- D. Adjusted R^2

B. RMSE

5. An R^2 value of 0.90 means:

- A. The model is 90% accurate.
- B. The slope is 0.90.
- C. Approximately 90% of the target variation is explained by the model.
- D. Every prediction has 10% error.

C. Approximately 90% of the target variation is explained by the model.



Thank you